

KinSync Feature Map:

what's built, what's next

KinSync quietly learns an elderly person's normal phone rhythm on their own phone, and alerts family when that rhythm goes silent. Raw activity data never leaves the phone. This map shows every planned feature against the IDP review schedule.

WE ARE HERE						
Review I	Review II	Review III	Review IV	Review V	Review VI	Report
17-21 Aug 2026	21-25 Sep 2026	12-16 Oct 2026	25-29 Jan 2027	8-12 Mar 2027	29 Mar-2 Apr 2027	2 Apr 2027
Problem, objectives, survey	~20% · design + first prototype	~30% · Phase 2	~50% · Phase 3	~80% · Phase 4	100% · Open House	Final report
5 marks · done	20 marks · done	10 marks · next	15 marks	25 marks	15 marks	10 marks

- MUST-HAVEneeded for the review demo
- NICE-TO-HAVEif time allows
- ANDROIDphone app · Sri Hasini
- BACKENDserver · Nithin
- DOCS

FR-2.1 requirement ID

Built and demoed

17 featuresPhase 1, shown at Review II (25 Sep 2026)

ANDROIDElder's phone

- ✓ Consent screen

Plain-language explanation before anything is collected.

FR-7.1 · early version
- ✓ Usage-access permission

Rationale screen, then a deep link to Android Settings.

FR-2.2
- ✓ Battery-optimisation exemption

Keeps collection alive; also asks for notification permission on Android 13+.

NFR-2
- ✓ Unlock and screen on/off capture

Runs in a foreground service so the phone maker's battery saver doesn't kill it.

FR-2.1

✓ **Restarts after reboot**

Only if the elder has already given consent.

NFR-2 · FR-7.1

✓ **On-phone database**

Events stored locally in Room. Nothing is uploaded.

FR-2.4 · NFR-1

✓ **Live event list**

Debug screen showing new unlock events within seconds.

FR-2.5 · precursor

✓ **Stop monitoring**

Revokes consent on the phone and stops collection.

FR-7.3 · partial

✓ **Backend reachability check**

Calls the server's health check on launch; HTTPS only.

NFR-4

✓ **Large-text theme**

Bigger type throughout for elderly users.

NFR-6

BACKEND

Server at kinsync.ddns.net

✓ **Hardened cloud server**

Hetzner VM, Ubuntu 26.04: key-only SSH, firewall, fail2ban, automatic security updates.

NFR-4

✓ **HTTPS with automatic certificates**

Caddy reverse proxy with Let's Encrypt.

NFR-4

✓ **API service**

FastAPI running as a system service that restarts itself on failure.

NFR-2

✓ **PostgreSQL database**

Reachable only from the server itself.

NFR-4

✓ **Health endpoints**

/health and /health/db, live.

Phase 1

✓ **Automated quality gate**

Lint, type checks, complexity limits and tests with at least 90% coverage.

NFR-7

DOCS

Project documentation

✓ **Specs, design, plan and runbooks**

Requirements, architecture, roadmap with commit numbers, operating procedures, traceability matrix.

kinsync-docs

Answers the Review II panel: "which features will be implemented?" The whole phase happens on the phone and nothing new leaves it. **Awaiting team approval.**

MUST-HAVE

App-usage pattern

Which apps were used and when, read every few minutes.

ANDROIDFR-2.2

Last moved

Uses the phone's low-power "significant motion" sensor and stores only the time. No step counting.

ANDROIDFR-2.7

Activity: still, walking, in vehicle

Google Activity Recognition, with a permission screen in plain language.

ANDROIDFR-2.3

"My day" timeline

All signals on one 24-hour view, with a day picker.

ANDROIDFR-2.5

Daily summary

First unlock, unlock count, screen time, top apps, last moved, time per activity.

ANDROIDFR-2.5

Consent screen lists every signal

The elder sees exactly what is collected before it starts.

ANDROIDFR-7.1

Tests for every new collector

Database tests on the device, logic tests on the computer.

ANDROIDNFR-7

Feature map for the panel

This page, kept up to date.

DOCS

NICE-TO-HAVE

Charging events

Plugged-in and unplugged times. No permission needed.

ANDROIDFR-2.8

Call activity

Number and times of calls only, never numbers or contacts. Needs a sensitive permission.

ANDROIDFR-2.8

30-day auto-delete

Raw events older than 30 days are removed from the phone.

ANDROIDNFR-1

First battery measurement

24 hours with and without collection, compared against the 5% target.

ANDROIDNFR-3

Draft of the Phase 3 API design

Data shapes for pairing, devices and heartbeat, so the backend is ready to start.

BACKEND

Remaining phases

24 features

Review IV to the Open House · Jan–Apr 2027

Phase 3 · Review IV

25–29 Jan 2027 · ~50% · core logic + first alert

Baseline learning

Rolling 14-day pattern, ready after 7 days.

MUST ANDROID FR-3

Deviation check + "Everything OK?" nudge

Asks the elder before alarming anyone.

MUST ANDROID FR-4.1–4.3

Elder / Caregiver role choice

MUST ANDROID FR-1.1

Pairing with a 6-digit code

Up to 3 caregivers per elder.

MUST ANDROID BACKEND FR-1.2–1.4

Device registration + secure tokens

MUST BACKEND NFR-4

Phase 4 · Review V

8–12 Mar 2027 · ~80% · caregiver side + validation

Caregiver dashboard

Status for each elder: normal, nudged or escalated.

MUST ANDROID FR-6.1

Acknowledge and resolve alerts

MUST ANDROID BACKEND FR-6.3 · FR-4.6

Activity trend for caregivers

Aggregated only: no timestamps or app names.

NICE ANDROID FR-6.2

Transparency screen

Everything collected, in plain language.

MUST ANDROID FR-2.5

Pause / travel mode

MUST ANDROID FR-2.6

Phase 5 · Open House + Report

29 Mar–2 Apr 2027 · 100%

Polish and accessibility pass

MUST ANDROID NFR-6

Performance evaluation

Alert delay, false-alarm rate, battery.

MUST G1 · G3

Open House demo kit

Two phones, live alert, backup video.

MUST

Final report

In the prescribed format.

MUST DOCS

SMS fallback for caregivers

NICE BACKEND FR-5.3

Server database tables

Elders, caregivers, pairings, heartbeats, alerts. No raw activity table.

MUST **BACKEND** FR-7.2

Heartbeat from the phone

Sends only "I'm OK" plus the expected check-in window.

MUST **ANDROID** **BACKEND** FR-7.2

Dead-man's switch

The server raises an alert when a heartbeat is missed, even if the phone is off.

MUST **BACKEND** FR-4.4

Push alert to caregivers

Firebase Cloud Messaging.

MUST **BACKEND** FR-4.5 · FR-5

Nightly database backups

Needed before real data exists.

MUST **BACKEND**

View and remove caregivers; revoking consent unpairs everyone

MUST **ANDROID** **BACKEND** FR-1.5 · FR-7.3

Pilot users and threshold tuning

Fewer false alarms.

MUST G3

Battery target: under 5% extra a day

MUST **ANDROID** NFR-3

Test plan and results

End-to-end and reliability tests.

MUST NFR-2

The privacy promise

- Raw data (unlocks, app names, motion, calls) stays on the phone.
- Only an "I'm OK" heartbeat and the expected check-in window are sent to the server.
- The server has no table that could hold raw activity.
- No GPS location, ever.

Out of scope for version 1

- Fall detection and medical diagnosis
- Step counting and continuous GPS tracking
- Wearables, iOS, audio or video monitoring
- Multiple languages; more than 3 caregivers per elder

Where the details live

- github.com/nithinvin/kinsync-docs: specs, design, roadmap
- github.com/nithinvin/kinsync-android: phone app
- github.com/nithinvin/kinsync-api: server

Phase 2 and later are plans. Priorities may change after each review.